CAPSTONE · AI FOR GOOD · WEEK 2 · READING 2.2

From stakeholders *to systems*

Sustainability systems, the Doughnut, and the data behind the energy transition.

Dr. Jan Erik Meidell · SUMAS

WHERE DID YOUR TEAM LAND?

Three questions after the team presentations

1. **Where do you agree?** Find two claims every member of the team would accept, about Ørsted, about Eni, or about the growth of wind.
2. **Where do you contradict each other?** Name one tension: where the data lead's numbers clash with someone's scenario, or with a feedback loop.
3. **What is still missing?** Which question did nobody answer well? That gap is your team's next research priority.

Good consulting synthesises partial views; it does not just list them.

THE CONTEXT · RAWORTH (2017)

We chose the wrong goal

"We have economies that need to grow whether or not they make us thrive. What we need are economies that make us thrive whether or not they grow."

Kate Raworth, TED talk, 2018

70+ years

of GDP as the dominant policy goal, since the 1940s. Raworth's image: economics lost sight of its goals, and GDP moved into the empty nest like a cuckoo.

Planetary boundaries crossed

4 of 9 when Raworth wrote; 6 of 9 by 2023 (Richardson et al., *Science Advances*); 7 of 9 in the 2025 Planetary Health Check, which added ocean acidification.

THE DOUGHNUT · A 21ST-CENTURY COMPASS

Between shortfall and overshoot

The social foundation · inner ring

Twelve essentials every person needs: food, water, energy, health, education, housing, income and work, political voice, gender equality, social equity, networks, peace and justice. Below it: *shortfall*.

The ecological ceiling · outer ring

Nine planetary boundaries: climate change, biodiversity loss, land conversion, freshwater, ozone layer, ocean acidification, chemical pollution, air pollution, nitrogen and phosphorus. Beyond it: *overshoot*.

Progress is not an ever-rising GDP line. It is humanity thriving in the safe and just space between the two rings.

APPLYING THE DOUGHNUT

Where European wind sits

Pressure on the ceiling

Rare earths and steel in turbines · cables and seabed impacts offshore · land use onshore · blades at end of life, not yet solved

Building the foundation

Affordable energy for households · a just transition for fossil-fuel workers · energy security for smaller member states · local acceptance of projects

Where Ørsted sits

A business model largely inside the safe space; its supply chain pushes against the ceiling; offshore projects affect marine biodiversity. Is it regenerative by design?

Where Eni sits

A core fossil business that overshoots the ceiling; Plenitude tries to rebalance, but at what pace? Gas and LNG against renewables growth: a direct tension. Can one company straddle both sides?

THE KEY MOVE

From a stakeholder map to a system

1 · Inventory

Who are the actors?

Ørsted, Eni, the European Commission, grid operators, banks, households, NGOs, regulators. Most Week 1 maps stop here.

2 · Interactions

How do they influence each other?

Regulation → investment → price signals → policy revision → new regulation. Systems analysts need to reach this level.

3 · Dynamics

What are the feedback loops?

Reinforcing: falling LCOE → more investment → more scale → lower LCOE. Balancing: permitting bottlenecks limit build-out despite falling costs. This is where strategy lives.

Raworth: the ecological ceiling is itself a feedback constraint, a balancing loop that caps economic activity whether we design for it or not.

WORKSHOP · DATA SCAVENGER HUNT

Your role has a data question

WHO	THE QUESTION: FIND IT IN TWO SOURCES, AND CHECK THEY AGREE	WHERE TO LOOK
Data lead	What share of EU electricity came from wind in 2024?	Eurostat · Ember · IEA
Model lead	The IEA's projection for European offshore wind in 2040, under each scenario	IEA WEO · WindEurope · ENTSO-E TYNDP
Ørsted	Ørsted's return on capital in offshore wind, 2022 against 2023	Ørsted annual reports
Eni	What share of Eni's capex went to Plenitude and renewables? The 2030 target?	Eni capital markets update · Plenitude reports
Coordinator	The rate of wind curtailment in Germany and Denmark, and where that data lives	Bundesnetzagentur / SMARD · Energinet · ACER
Editor	Which of Raworth's five Doughnut factors (population, distribution, aspiration, technology, governance) shows most in the EU wind case?	Raworth, chapter 1, and the rest of this table

SOURCE QUALITY

Know what your data actually is

Primary data

Company filings (annual reports, CDP), official statistics (Eurostat, Bundesnetzagentur), peer-reviewed datasets.
✓ Traceable, audited or reviewed.
⚠ One or two years' lag; some figures self-reported.

Aggregated statistics

IRENA, IEA, WindEurope, Ember: primary data compiled and cleaned for policy use.
✓ Widely cited by regulators.
⚠ Methods vary; "renewable" is defined differently.

Modelled projections

IEA scenarios, ENTSO-E TYNDP, BloombergNEF outlooks to 2040+.
✓ Essential for foresight.
⚠ Built on assumptions about policy, technology and behaviour: interrogate them.

Before citing a number: is it measured, compiled or modelled? Which year? Which geography? Which definition?

BEFORE WEEK 3

A one-page data memo, per person

1. Identify **three datasets** that answer your role's question from the scavenger hunt.
2. Open or download **at least one**. Show the key figure or table.
3. Write three sentences: what the data tells you, what it does **not** tell you, and what question stays open.

Week 3 is about AI tools for analysis: you need real data to bring into them. Come with data, leave with insight.

CLOSE

Thrive *in balance*

"Ours is the era of the planetary household, and the art of household management is needed more than ever for our common home."

Kate Raworth, *Doughnut Economics* (2017)